

What is CSS Masking?

CSS Masking allows you to hide or show portions of an element using a mask image or gradient. A mask uses luminance (brightness) values to determine visibility: black areas are fully transparent, white areas are fully opaque, and gray areas show variable transparency.

Masking is different from clipping (`clip-path`): clipping creates sharp geometric boundaries, while masking uses smooth gradients or images for sophisticated visual effects.

Browser Support: Chrome, Firefox, Safari, and Edge support CSS masking with `-webkit-` prefix for better compatibility. Use both prefixed and unprefixed versions in production.

Core Masking Concepts

How Masking Works

Masking uses luminance (brightness) values to control transparency:

- White (100%):** Fully opaque - content is completely visible
- Gray (50%):** Semi-transparent - content is partially visible
- Black (0%):** Fully transparent - content is completely hidden

Mask vs Opacity

Property Effect Use Case `opacity` Uniform transparency applied to entire element Fade effects, hover states `mask` Selective transparency - different areas can be hidden/shown Vignette effects, creative shapes, gradual fades `clip-path` Sharp geometric clipping boundary Polygons, circles, simple shapes

Mask vs Filter

`mask` controls visibility, while `filter` (like blur, brightness) modifies appearance. They work together nicely for advanced effects.

CSS Masking Methods

1. mask-image Property

Applies an image as the mask. Black areas hide content, white shows it.

Syntax: `mask-image: url(image.png);`

Supported Formats:

- PNG with alpha channel
- SVG images with masks
- Gradients (linear, radial, conic)

Examples:

```
mask-image: url('mask.png');
mask-image: linear-gradient(90deg, black, transparent);
mask-image: radial-gradient(circle, black 50%, transparent);
```

2. mask-size Property

Controls the size of the mask image (similar to background-size).

Syntax: `mask-size: <width> <height>;`

Values:

- auto (default)
- percentages: 50%, 100%
- keywords: cover, contain
- pixel values: 200px, 300px

Examples:

```
mask-size: 100% 100%;
mask-size: cover;
mask-size: 50px 50px;
```

3. mask-position Property

Positions the mask within the element (similar to background-position).

Syntax: `mask-position: <x> <y>;`

Values: Keywords (left, right, center, top, bottom) or percentages

Examples:

```
mask-position: center;
mask-position: top left;
mask-position: 50% 50%;
```

4. mask-repeat Property

Controls whether and how the mask repeats (similar to background-repeat).

Syntax: `mask-repeat: <value>;`

Values:

- no-repeat (default)
- repeat
- repeat-x
- repeat-y
- space
- round

5. mask-origin Property

Determines the reference box for the mask (border-box, padding-box, or content-box).

Syntax: `mask-origin: <value>;`

Values:

- border-box (default)
- padding-box
- content-box

6. mask-clip Property

Clips the mask to specific regions.

Syntax: `mask-clip: <value>;`

Values:

- border-box
- padding-box
- content-box
- fill-box
- stroke-box
- view-box

7. mask Shorthand

Combines all mask properties in one declaration.

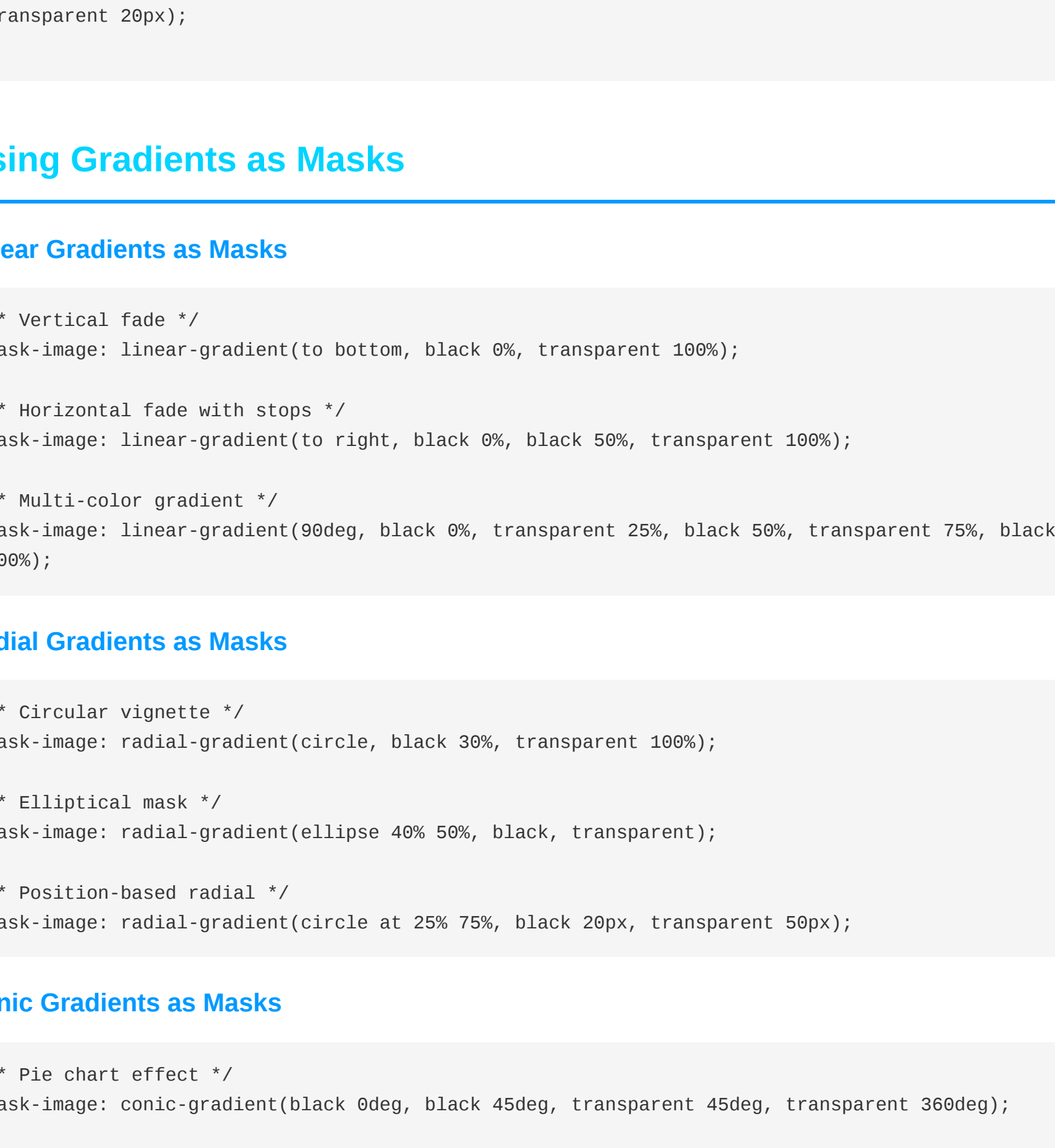
Syntax: `mask: [image] [position] / [size] [repeat] [origin] [clip];`

Examples:

```
mask: url('mask.png') center / contain no-repeat;
mask: linear-gradient(90deg, black, transparent) center / cover;
```

Interactive Masking Demos

Explore different masking techniques and their effects on the gradient background.



Common Use Cases & Patterns

Vignette Effect (Darkened Edges)

```
img {
  -webkit-mask-image: radial-gradient(circle at center, black 0%, black 60%, transparent 100%);
  mask-image: radial-gradient(circle at center, black 0%, black 60%, transparent 100%);
}
```

Fade Out to Transparent

```
.element {
  -webkit-mask-image: linear-gradient(180deg, black 0%, transparent 100%);
  mask-image: linear-gradient(180deg, black 0%, transparent 100%);
}
```

Directional Fade (Left to Right)

```
.element {
  -webkit-mask-image: linear-gradient(90deg, transparent 0%, black 25%, black 75%, transparent 100%);
  mask-image: linear-gradient(90deg, transparent 0%, black 25%, black 75%, transparent 100%);
}
```

Spotlight Effect

```
.spotlight {
  -webkit-mask-image: radial-gradient(circle 100px at center, black, transparent);
  mask-image: radial-gradient(circle 100px at center, black, transparent);
}
```

Animated Mask Reveal

```
@keyframes revealMask {
  0% { -webkit-mask-position: -200px 0; mask-position: -200px 0; }
  100% { -webkit-mask-position: 0 0; mask-position: 0 0; }
}

.element {
  -webkit-mask-image: linear-gradient(90deg, transparent 0%, black 50%, transparent 100%);
  mask-image: linear-gradient(90deg, transparent 0%, black 50%, transparent 100%);
  -webkit-mask-size: 200px 100%;
  mask-size: 200px 100%;
  animation: revealMask 3s infinite;
}
```

Image with PNG Mask

```
img {
  -webkit-mask-image: url('mask.png');
  mask-image: url('mask.png');
  -webkit-mask-size: cover;
  mask-size: cover;
}
```

Stripe Pattern Mask

```
.element {
  -webkit-mask-image: repeating-linear-gradient(90deg, black 0px, black 10px, transparent 10px, transparent 20px);
  mask-image: repeating-linear-gradient(90deg, black 0px, black 10px, transparent 10px, transparent 20px);
}
```

Using Gradients as Masks

Linear Gradients as Masks

```
/* Vertical fade */
mask-image: linear-gradient(to bottom, black 0%, transparent 100%);

/* Horizontal fade with stops */
mask-image: linear-gradient(to right, black 0%, black 50%, transparent 100%);

/* Multi-color gradient */
mask-image: linear-gradient(90deg, black 0%, transparent 25%, black 50%, transparent 75%, black 100%);
```

Radial Gradients as Masks

```
/* Circular vignette */
mask-image: radial-gradient(circle, black 30%, transparent 100%);

/* Elliptical mask */
mask-image: radial-gradient(ellipse 40% 50%, black, transparent);

/* Position-based radial */
mask-image: radial-gradient(circle at 25% 75%, black 20px, transparent 50px);
```

Conic Gradients as Masks

```
/* Pie chart effect */
mask-image: conic-gradient(black 0deg, black 45deg, transparent 45deg, transparent 360deg);

/* Radial segments */
mask-image: conic-gradient(from 0deg, black 0deg, black 90deg, transparent 90deg, transparent 180deg, black 180deg);
```

Advanced Masking Techniques

Animated Mask Transitions

```
.element {
  -webkit-mask-image: radial-gradient(circle, black 0%, transparent 100%);
  mask-image: radial-gradient(circle, black 0%, transparent 100%);
  transition: -webkit-mask-image 0.5s ease;
  transition: mask-image 0.5s ease;
}

.element:hover {
  -webkit-mask-image: radial-gradient(circle, black 0%, transparent 50%);
  mask-image: radial-gradient(circle, black 0%, transparent 50%);
}
```

Multiple Masks

```
/* Stack multiple masks */
.element {
  -webkit-mask-image:
    linear-gradient(90deg, black, transparent),
    radial-gradient(circle, transparent 30%, black 70%);
  mask-image:
    linear-gradient(90deg, black, transparent),
    radial-gradient(circle, transparent 30%, black 70%);
  -webkit-mask-composite: source-in;
  mask-composite: intersect;
}
```

SVG Mask Reference

```
/* Reference SVG mask defined inline or external */
img {
  -webkit-mask-image: url('#my-mask-id');
  mask-image: url('#my-mask-id');
}
```

Dynamic Mask Size

```
.element {
  -webkit-mask-image: radial-gradient(circle, black, transparent);
  mask-image: radial-gradient(circle, black, transparent);
  -webkit-mask-size: 200px 200px;
  mask-size: 200px 200px;
  -webkit-mask-position: center;
  mask-position: center;
}
```

Browser Compatibility & Prefixes

Chrome/Edge: Full support, use both `-webkit-` and unprefixed versions.

Firefox: Full support since version 53. Use unprefixed properties.

Safari: Support with `-webkit-` prefix. Always include prefixed versions.

IE 11: Does NOT support CSS masks. Provide fallbacks or graceful degradation.

Vendor Prefix Strategy

```
/* Always include both for maximum compatibility */
.element {
  /* Webkit (Chrome, Safari, Edge Chromium) */
  -webkit-mask-image: linear-gradient(90deg, black, transparent);
  -webkit-mask-size: cover;
  -webkit-mask-position: center;

  /* Standard (Firefox, Edge) */
  mask-image: linear-gradient(90deg, black, transparent);
  mask-size: cover;
  mask-position: center;
}
```

Best Practices

- Always use vendor prefixes:** Include both `-webkit-` and unprefixed versions for broad browser support.
- Test performance:** Complex mask gradients can impact performance. Test on slower devices.
- Test mask images:** Ensure PNG masks have proper alpha channels. Use tools like ImageMagick to verify.
- Combine with opacity:** Use `opacity` for overall transparency, `mask` for selective visibility.
- Fallback content:** For IE 11 compatibility, provide alternative styling or images.
- Avoid complex masks on low-end devices:** Rendering multiple or complex masks can cause lag on mobile.
- Use gradients over images when possible:** Gradients are lighter weight than image masks.
- Document mask intent:** Add comments explaining what each mask is supposed to do for maintainability.

Mask vs clip-path: When to Use Each

Aspect	mask	clip-path
Boundary Type	Smooth, gradient-based	Sharp, geometric
Transparency	Variable (black to white)	Binary (hidden or shown)
Use Case	Vignette, fade, spotlight, artistic	Circles, polygons, triangles
Performance	Medium (especially with gradients)	High (very efficient)
Browser Support	Chrome, Firefox, Safari	All modern browsers
Animation	Smooth gradient animations possible	Sharp boundary animations

Common Mistakes to Avoid

- Forgetting vendor prefixes:** Many browsers still require `-webkit-` prefix. Don't omit it.
- Using PNG without alpha channel:** Masks need transparency. Ensure PNG has proper alpha channel.
- Assuming mask-image works everywhere:** IE 11 and older browsers don't support it. Always provide fallbacks.
- Not testing mask-size and mask-position:** These properties are necessary for proper mask placement.
- Complex masks on many elements:** This can cause performance issues. Use sparingly.
- Forgetting mask-repeat:** Without specifying, masks may repeat unexpectedly.
- Using only gradients without contrast:** Low-contrast gradients may not mask effectively. Use clear black-to-white transitions.
- Not considering accessibility:** Masked content should still be accessible to screen readers.

Quick Reference - Common Patterns

Template: Basic Mask Setup

```
.element {
  /* Webkit browsers */
  -webkit-mask-image: linear-gradient(to bottom, black, transparent);
  -webkit-mask-size: 100% 100%;
  -webkit-mask-position: 0 0;

  /* Standard */
  mask-image: linear-gradient(to bottom, black, transparent);
  mask-size: 100% 100%;
  mask-position: 0 0;
}
```

Template: Vignette Effect

```
.vignette {
  -webkit-mask-image: radial-gradient(circle at center, black 0%, black 50%, transparent 100%);
  mask-image: radial-gradient(circle at center, black 0%, black 50%, transparent 100%);
}
```

Template: Fade Bottom

```
.fade-bottom {
  -webkit-mask-image: linear-gradient(180deg, black 0%, transparent 100%);
  mask-image: linear-gradient(180deg, black 0%, transparent 100%);
}
```

Template: Animated Reveal

```
@keyframes reveal {
  0% { -webkit-mask-position: 200% 0; mask-position: 200% 0; }
  100% { -webkit-mask-position: -200% 0; mask-position: -200% 0; }
}

.element {
  -webkit-mask-image: linear-gradient(90deg, transparent 0%, black 50%, transparent 100%);
  mask-image: linear-gradient(90deg, transparent 0%, black 50%, transparent 100%);
  -webkit-mask-size: 200% 100%;
  mask-size: 200% 100%;
  animation: reveal 3s infinite;
}
```